Defense 1 pipeline is working end to end

NON-WORKING FEATURES

None identified in the final Docker rehearsal.

High, medium, and clean ingestion; Teams alerts; Data Docs; API predictions; history; and no-new-data skipping were verified. The main release remains intentionally pending.

62

automated tests passed

5/5

ingestion tasks passed

37

scheduled predictions saved

Each member owns a complete pipeline layer

The implementation can be presented as one connected flow: data quality -> model -> API -> web demo.

Fateh Cherrad

Data quality & integration

- Data splitting and corruption generation
- Great Expectations and Data Docs
- Statistics, Teams alerts, final rehearsal

Defense focus
Ingestion lifecycle and integration

Yassine

Model & Airflow platform

- Baseline and upgraded model
- Training metrics and model tests
- Airflow and Docker backend setup

Defense focus
Model choices and orchestration

Ashish

API & persistence

- FastAPI schemas and endpoints
- PostgreSQL and SQLAlchemy persistence
- Prediction history and prediction DAG

Defense focus
API contracts and stored results

Akash

Web application & demo

- Streamlit single and batch prediction
- User-friendly feature inputs
- Past predictions and Streamlit Docker

Defense focus
Live prediction workflow